

Project 1: House Price Prediction Summary

Part 1: California Housing Dataset

Initial Exploration and Preprocessing:

The categorical `ocean_proximity` feature was converted to a numerical format via One-Hot Encoding, Dropping the first encoded column was necessary to prevent the dummy variable trap (multicollinearity). Label Encoding was rejected as an alternative because it gives different categories more power.

Imputation to handle Missing values was done using a K-Nearest Neighbors (KNN) Imputer (`n_neighbors=5`). This necessary step estimates missing data based on feature similarity rather than a simplistic overall mean, avoiding the loss of dropping incomplete rows.

The `median_house_value` target variable contained an artificial cap at \$500,001, which was removed. Leaving these capped outliers creates an unnatural horizontal boundary that skews the regression plane.

Heavily right-skewed features like `total_rooms` and `population` were normalized using a logarithmic transformation to satisfy linear regression assumptions. While a Box-Cox transformation was considered, log transformation proved sufficient, leaving the data untransformed was rejected as it severely degrades accuracy.

Because the features operated on vastly different scales, `StandardScaler` was applied to prevent large-magnitude variables from biasing and to ensure efficient gradient convergence, Min-Max scaling was rejected due to its high sensitivity to outliers.

Feature Engineering and Model Selection:

Baseline raw variables (such as *total_rooms*, *households*, and *population*) exhibited high correlation; by adding a ratio feature, we eliminated severe multicollinearity, demonstrating a significant boost in predictive accuracy, ultimately selected as the best overall model, which was using L2 penalty which was better than L1 or OLS.

Part 2: Boston Housing Dataset

Initial Exploration and Preprocessing:

An initial exploration revealed scaling issues and a complex mix of data distributions. Missing values in the `rm` column were handled by simply dropping the 5 incomplete rows. Imputation was not necessary as 5 rows is too small.

The `medv` target variable contained an artificial price cap at exactly 50.0, which was removed. Leaving these capped outliers creates an unnatural horizontal boundary that skews the regression line at the top end, strict removal provided better linear relationships than capping at a calculated percentile.

Heavily right-skewed features like `crim`, `zn`, `lstat`, and `dis` were normalized using a logarithmic transformation to satisfy linear regression assumptions. For the left-skewed data, a square transformation was initially tested; however, this approach was ultimately abandoned because it resulted in lower accuracy when testing.

Because the features operated on vastly different scales, `StandardScaler` was applied to prevent large-magnitude variables from biasing and to ensure efficient gradient convergence, `Min-Max` scaling was rejected due to its high sensitivity to outliers.

Feature Engineering and Model Selection:

Baseline raw variables (such as `tax` and `rad`) exhibited extreme positive correlation (> 0.90); by dropping the `tax` variable entirely, we eliminated severe multicollinearity and satisfied independence assumptions, demonstrating a boost in validation metrics.

Ultimately selected as the best overall model, which was using L2 penalty (Ridge Regression) which was better than L1 or OLS.

Analyzing the coefficients provided critical insights into feature importance and model alignment. Furthermore, regression assumptions were verified; the residual scatter plot confirmed homoscedasticity, and the residual histogram displayed a roughly normal distribution, validating our inferences.